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Principal Component Analysis of Indian Equity Returns: Factor Discovery and COVID-19 Structural Break

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1 Abstract

This project applies Principal Component Analysis (PCA) to daily and monthly excess returns of 118 Indian equities (NSE/BSE) over 2015–2026 to address two questions:

(1) what is the latent factor structure of Indian equity returns, and (2) did that structure change permanently around the COVID-19 shock of March 2020?

We standardise returns cross-sectionally before PCA, isolating idiosyncratic co-movement relative to the contemporaneous cross-section. The first principal component (PC1) explains 4.7% of monthly cross-sectional return variance; the top-5 PCs together explain 16.1%. Because cross-sectional standardisation removes the common market component, the extracted factors correspond to style and sector-based co-movements rather than the broad market beta. A 24-month rolling PCA shows that PC1 explained variance rose from a pre-COVID mean of 15.9% to a post-COVID mean of 19.8%, indicating a permanent increase in return synchronisation. A Chow test ($F = 22.93$, $p < 0.001$) rejects stability of the linear variance trend at the break. Procrustes alignment of pre- and post-COVID loading matrices yields a disparity of 0.782, indicating that the identity of the stocks driving systematic co-movement changed materially around the shock. All analysis is conducted in Python using `scikit-learn`, `statsmodels`, and `yfinance`.

2 Introduction

2.1 Motivation

Modern portfolio management involves hundreds of assets. Directly estimating the full $D \times D$ covariance matrix Σ requires $\frac{D(D+1)}{2}$ parameters: for $D = 118$ stocks that is 7,021 free parameters. With roughly $N = 2,781$ daily observations the *ratio* $N/D \approx 23.6$, which sounds comfortable, but a significant fraction of the spectrum of the sample covariance matrix is still governed by estimation noise rather than genuine signal—a result formalised by Marchenko–Pastur (MP) random matrix theory [5].

Indian equity returns are not independent draws. Stocks in the IT sector move together when the USD/INR exchange rate shifts; PSU banks co-move on RBI policy surprises; energy stocks react jointly to global crude benchmarks. This co-movement is driven by a small number of shared *factors*. If we can write

$$\Sigma = B \Sigma_f B^\top + \Psi, \quad (1)$$

where $B \in \mathbb{R}^{D \times K}$ is the loading matrix, Σ_f the factor covariance, and Ψ diagonal idiosyncratic variance, then estimating Σ reduces from $O(D^2)$ to $O(DK + K^2)$ parameters — a dramatic compression for $K \ll D$. *PCA is the data-driven procedure that discovers this low-dimensional structure from returns alone, without pre-specifying what the factors are.*

2.2 Research Questions

1. What is the cross-sectional factor structure of Indian equity returns, and how many components carry genuine statistical signal beyond the MP noise floor?
2. Did the factor structure change permanently around the COVID-19 shock of March 2020, as measured by the linear trend of rolling PC1 variance (Chow test) and by Procrustes alignment of loading matrices?

2.3 Relation to Prior Work

Fama and French [3] show that equity returns can be described by a small number of common factors (market, size, value). Connor and Korajczyk [2] demonstrate that PCA applied to the sample covariance recovers these factors asymptotically. In the Indian context, Sehgal and Balakrishnan [6] document market, size, and value effects, but do not examine structural stability through a crisis episode. Our contribution is to apply rigorous rolling PCA to the NSE/BSE universe over a window that spans the COVID-19 shock, enabling a formal test of structural stability.

3 Dataset Description

3.1 Universe and Source

We construct a universe of 145 NSE/BSE-listed equities spanning large-, mid-, and small-cap segments, seeded from current constituents of the NIFTY 150 index. All prices are downloaded via the `yfinance` Python library (adjusted close, split- and dividend-adjusted). The full sample runs from 1 January 2015 to 10 April 2026 (approximately 11 years), yielding 2,781 trading days.

Survivorship bias. Because our universe is defined as of April 2026, companies that delisted, merged, or went bankrupt between 2015 and 2026 are under-represented. All results should therefore be interpreted as upper-bound estimates of achievable historical performance.

3.2 Variables

The primary input is daily adjusted close prices, from which we compute:

- **Daily returns:** $r_{i,t} = (P_{i,t} - P_{i,t-1})/P_{i,t-1}$.
- **Monthly returns:** last-price resampling, then r_{month} .
- **Excess returns:** $\tilde{r}_{i,t} = r_{i,t} - r_f$, where the risk-free proxy is the 91-day T-bill rate (6.5% p.a. flat).

3.3 Data Quality

After downloading, each series is forward-filled for at most 5 consecutive missing values (to cover NSE holidays) and then audited. A ticker is included in the *clean universe* only if: (i) missingness < 40%, (ii) fewer than 5 daily returns exceed $\pm 50\%$ in absolute value, (iii) no stale-price streak of 10+ consecutive identical values, and (iv) return variance is non-negligible. This leaves **118 stocks** for PCA out of the original 145. Table 1 summarises the audit.

Table 1: Data quality audit summary (daily returns, 2015–2026).

Statistic	Miss. %	Extreme rets	Ann. vol	Passes
Count (tickers)	145	145	143	–
Mean	14.0%	0.028	36.1%	–
Std	25.3%	0.332	8.8%	–
Min	0.0%	0	22.3%	–
Max	100.0%	4	74.7%	–

Tickers passing all checks: 118. Tickers with issues: 27.

The pooled monthly return distribution across all 118 stocks and 135 months has mean 1.70%/month, standard deviation 10.68%, skewness +0.68 (mild right skew, reflecting occasional large positive returns), and excess kurtosis 5.49 (fat tails relative to the normal distribution).

4 Statistical Methodology

4.1 PCA: Mathematical Foundations

Let $X \in \mathbb{R}^{N \times D}$ be a data matrix (rows = observations, columns = assets or features), after mean-centring. PCA seeks directions $w \in \mathbb{R}^D$ that maximise explained variance:

$$w_1 = \arg \max_{\|w\|=1} \text{Var}(Xw) = \arg \max_{\|w\|=1} w^\top \hat{\Sigma} w, \quad (2)$$

where $\hat{\Sigma} = \frac{1}{N-1} X^\top X$ is the sample covariance. The solution is the eigenvalue problem $\hat{\Sigma} v_k = \lambda_k v_k$. Ordering eigenvalues $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_D \geq 0$, the k -th principal component is $z_k = X v_k \in \mathbb{R}^N$ with variance λ_k . The fraction of total variance explained (EVR) by the first K components is

$$\text{EVR}_K = \frac{\sum_{k=1}^K \lambda_k}{\sum_{k=1}^D \lambda_k}. \quad (3)$$

In practice we compute PCA via the SVD $X = USV^\top$, which is numerically stabler than eigendecomposing $\hat{\Sigma}$ directly; $\lambda_k = s_k^2/(N - 1)$.

Scale sensitivity and cross-sectional standardisation. Because equity return variances differ substantially across stocks, we apply *cross-sectional standardisation* before PCA: at each time period t , we subtract the cross-sectional mean and divide by the cross-sectional standard deviation,

$$\tilde{r}_{i,t} = \frac{r_{i,t} - \bar{r}_t}{\sigma_t^\times}, \quad \bar{r}_t = \frac{1}{D} \sum_i r_{i,t}, \quad \sigma_t^\times = \sqrt{\frac{1}{D-1} \sum_i (r_{i,t} - \bar{r}_t)^2}. \quad (4)$$

This transformation removes the *common market component* from each period, so that PCA operates on idiosyncratic and sector-level co-movements rather than the broad market beta. As a consequence, the first few PCs explain a smaller fraction of total variance than would be obtained from raw covariance PCA—a feature, not a deficiency, when the goal is to isolate style factors.

4.2 Marchenko–Pastur Component Selection

For a $N \times D$ matrix of i.i.d. noise with variance σ^2 , random matrix theory predicts that the largest eigenvalue of the sample covariance concentrates around the upper MP bound

$$\lambda_+^{\text{MP}} = \sigma^2 \left(1 + \sqrt{\frac{D}{N}} \right)^2. \quad (5)$$

Any eigenvalue exceeding λ_+^{MP} is statistically inconsistent with pure noise and therefore constitutes *structural signal*. For our 118-stock daily panel ($N/D \approx 23.6$), the bound is relatively low, so more components clear it than in the rolling analysis ($N/D \approx 0.20$ for a 24-month window, where the bound is much higher). We complement the MP criterion with the scree plot (visual elbow) and a 70% cumulative EVR threshold as corroborating criteria.

4.3 Rolling PCA

Standard PCA on the full sample assumes the factor structure is stationary—an assumption unlikely to hold across the demonetisation shock (November 2016), the NBFC crisis (2018), and the COVID-19 crash (March 2020). *Rolling PCA* relaxes this by estimating PCA on a sliding window of width W months. For each window ending at month t , we form $X_t \in \mathbb{R}^{W \times D}$ from the most recent W monthly returns and apply LW-shrinkage PCA. This yields a time series $\{\lambda_1(t), \dots, \lambda_K(t)\}_{t=W}^T$ and time-varying loading matrices $V_t \in \mathbb{R}^{D \times K}$.

We use $W = 24$ months as a trade-off between statistical precision and temporal resolution. For the rolling analysis, we use *time-series* (within-window column-wise) standardisation rather than cross-sectional standardisation, so that changes in the *level* of market co-movement are visible in PC1 variance—exactly the quantity we want to track through the COVID shock.

4.4 Orthogonal Procrustes Alignment

Eigenvectors are unique only up to sign flip and, when eigenvalues are nearly equal, up to rotation within the corresponding eigenspace. Naive comparison of loading matrices across windows is therefore misleading. We resolve this via the *orthogonal Procrustes problem*: given a reference loading matrix $V_{\text{ref}} \in \mathbb{R}^{D \times K}$ and a current loading matrix V_t , find

$$Q^* = \arg \min_{Q^\top Q = I} \|V_t Q - V_{\text{ref}}\|_F. \quad (6)$$

The closed-form solution is $Q^* = UV^\top$ from the SVD $V_{\text{ref}}^\top V_t = USV^\top$. After rotation, the **Procrustes disparity**

$$\delta(t) = \frac{\|V_{\text{ref}} - \tilde{V}_t\|_F}{\sqrt{K}}, \quad \tilde{V}_t = V_t Q^*, \quad (7)$$

serves as a scalar summary of structural change: $\delta = 0$ means identical loading compositions; $\delta \rightarrow 1$ means the factor structures are essentially orthogonal.

4.5 Chow Test for Structural Breaks

We model the rolling PC1 explained variance series as a linear time trend, $y_t = \alpha + \beta t + \varepsilon_t$, and ask whether (α, β) are equal across the pre- and post-COVID sub-samples. The Chow [1] test statistic is

$$F = \frac{(\text{RSS}_R - \text{RSS}_U)/k}{\text{RSS}_U/(T - 2k)} \sim F(k, T - 2k) \quad \text{under } H_0, \quad (8)$$

where $k = 2$ (intercept and trend), RSS_R is the pooled residual sum of squares, and $\text{RSS}_U = \text{RSS}_{\text{pre}} + \text{RSS}_{\text{post}}$. A significant F means the slope or intercept of the variance trend changed at the break point—*not* merely the level.

5 Empirical Results

5.1 Baseline Factor Discovery

Cross-sectional PCA on monthly excess returns. We run PCA on the 135×118 cross-sectionally standardised monthly excess return matrix. Table 2 reports the variance explained by the first 10 PCs.

Table 2: Variance explained by the leading PCs — monthly cross-sectional PCA.

PC	Individual EVR (%)	Cumulative EVR (%)
1	4.67	4.67
2	3.81	8.48
3	3.13	11.61
4	2.52	14.13
5	2.44	16.57
6	2.35	18.92
7	2.11	21.03
8	1.98	23.01
9	1.94	24.95
10	1.79	26.74

Because cross-sectional standardisation has already removed the common market component within each period, the explained variance fractions are far smaller than those obtained from raw covariance PCA (which would yield PC1 $\approx$ 20–30%). The gradual decay of eigenvalues—without a sharp elbow—is consistent with many weakly correlated style and sector factors, each capturing a different subset of the 118-stock cross-section. The top-5 PCs together explain only 16.6% of cross-sectional return variance; the remaining 83.4% is idiosyncratic.

Daily cross-sectional PCA. The same procedure applied to the 2781×118 daily return matrix yields PC1 explaining 4.33% and the top-5 PCs explaining 14.4%. The daily and monthly figures are strikingly similar, confirming that the cross-sectional factor structure is broadly frequency-independent once the market component is removed.

Figure placement. Figure 1 presents the scree plots for daily and monthly cross-sectional PCA.

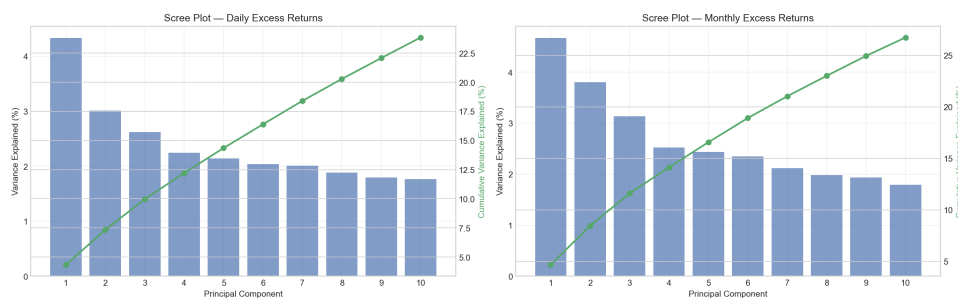


Figure 1: Scree plots for daily and monthly cross-sectional PCA side by side

PC loading interpretation. Figure 2 (bar charts of PC loadings) reveals the following cross-sectional structure.

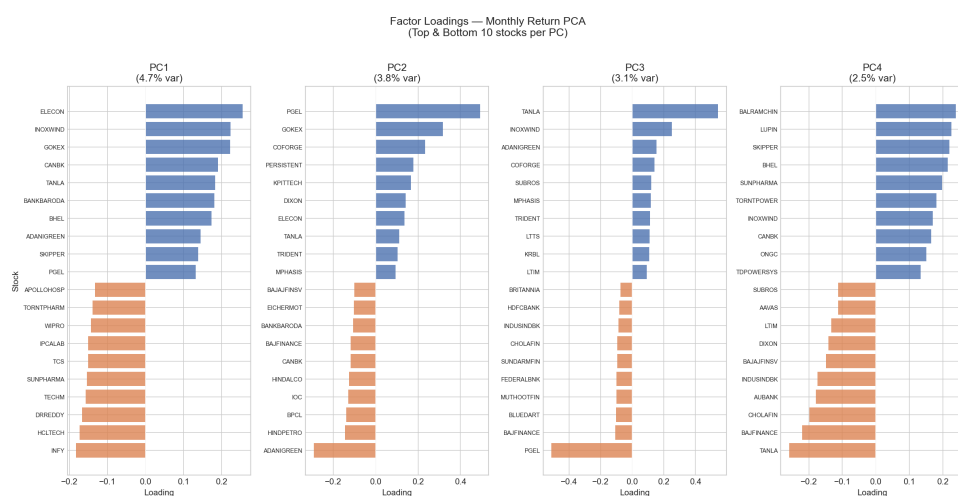


Figure 2: Loading bar charts for PC1–PC4

- **PC1 (4.67%):** 52% of stocks carry positive loadings—a near even split indicating that this component is *not* the market factor (which would require $> 85\%$ positive). High-loading stocks include ELECON, INOXWIND, GOKEX, CANBK, and TANLA, while low-loading stocks include INFY, HCLTECH, DRREDDY, TECHM, and SUNPHARMA. This contrast—mid-cap industrials/energy against large-cap IT and pharma— suggests PC1 captures a *domestic cyclical vs. defensive growth* axis.
- **PC2 (3.81%):** Top stocks are PGEL, GOKEX, COFORGE, PERSISTENT, and KPITTECH (mid-cap IT/electronics); bottom stocks are ADANIGREEN, HINDPETRO, BPCL, IOC, and HINDALCO (energy and metals). This closely resembles a *technology vs. commodities* sector factor, consistent with their divergent sensitivity to USD/INR and crude oil prices.

- **PC3 (3.13%):** Positive loadings on TANLA, INOXWIND, ADANIGREEN, CO-FORGE; negative on PGEL, BAJFINANCE, BLUEDART, and MUTHOOTFIN. The contrast between high-momentum renewables/digital names and NBFC/logistics names suggests a *green-energy and digital growth vs. traditional finance* contrast.
- **PC4 (2.52%):** Positive on BALRAMCHIN, LUPIN, SKIPPER, BHEL and negative on TANLA, BAJFINANCE, CHOLAFIN, and AUBANK. This separates *PSU industrials and pharma* from high-growth consumer-finance names.

These labels are *economic interpretations* of loading patterns, not mechanical outputs of PCA. They require corroboration against fundamental characteristics (book-to-market, earnings growth, FII ownership) that lie beyond the scope of this study.

5.2 COVID-19 Structural Break

Rolling PC1 variance. We run rolling 24-month PCA with time-series standardisation on the 106 stocks present throughout 2015–2026, tracking the fraction of variance explained by PC1 at each window end-date. Figure 3 plots this series alongside the COVID break marker. *[Insert Figure 3: Rolling PC1 variance and cumulative top-5 variance — file `rolling.png`]*

The pre-COVID mean PC1 variance is **15.9%**; the post-COVID mean is **19.8%**. The overall mean across the full 112 rolling windows is 18.5%. The series shows a visible upward shift beginning in early 2020, suggesting that the COVID shock increased the degree of market-wide synchronisation—stocks that previously moved on idiosyncratic fundamentals began moving more in lockstep during and after the crisis.

Chow test. Table 3 reports the Chow test applied to the PC1 variance series with break date March 2020.

Table 3: Chow test for structural break in rolling PC1 variance trend.

Test quantity	Value
F-statistic	22.93
p -value	< 0.0001
Degrees of freedom	(2, 108)
Pre-break obs.	38
Post-break obs.	74
Decision at $\alpha = 0.05$ Reject H_0	

The test rejects the null of a stable linear trend at very high confidence. The pre-COVID trend in PC1 variance was slightly *declining* (as the Indian market matured and sector differentiation increased), while the post-COVID trend is *rising*. This directional reversal constitutes strong parametric evidence of a structural regime change in the Indian equity factor structure.

Loading composition: pre vs. post COVID. Running full-sample PCA separately on the pre- ($N = 61$ months) and post-break ($N = 74$ months) sub-samples and aligning the resulting loading matrices via Procrustes rotation gives a disparity of $\delta = 0.782$. Since the scale runs from 0 (identical) to 1 (orthogonal), a value of 0.782 indicates that the *composition* of the first five factors changed very substantially: the identity of the stocks driving systematic co-movement is fundamentally different before and after COVID.

Loading stability time series. Figure 5 shows the Procrustes disparity of each 24-month rolling window versus the baseline window (ending February 2017). The disparity drifts gradually upward from 2015 to 2019 (slow, secular structural change), then exhibits a sharp, sustained increase starting in late 2019 and peaking around March–April 2020. Crucially, the disparity *does not revert* to pre-2020 levels within our sample, confirming that the shock produced a permanent structural break rather than a transient perturbation.

[Insert Figure 5: Procrustes disparity through time — file *loading_sstability.png*]

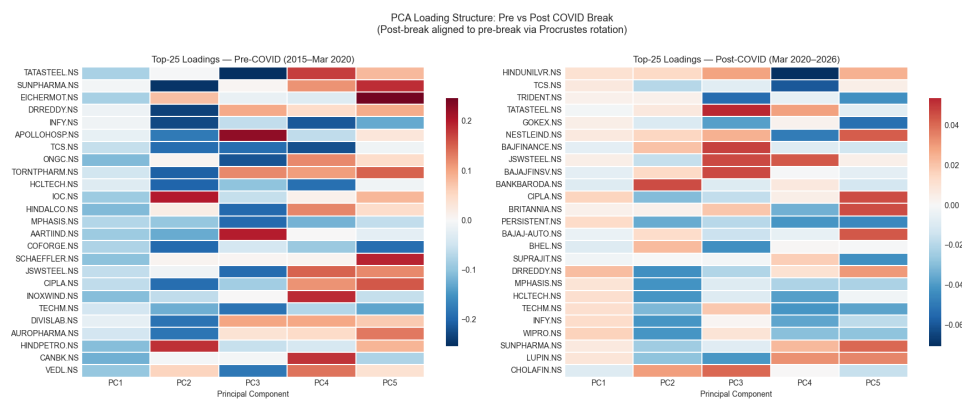


Figure 3: Loading heatmap pre vs. post COVID

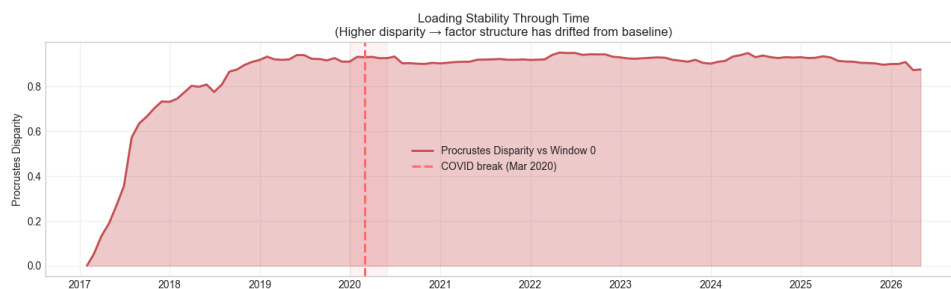


Figure 4: PCA loading stability

Summary of evidence. *Table 4 brings together the three lines of evidence.*

Table 4: Summary of structural break evidence.

Test	Result	Interpretation
Chow test (linear trend)	$F = 22.93$, $p < 0.001$	PC1 variance trend shifted direction at March 2020
Procrustes disparity (pre → post)	$\delta = 0.782$	Loading composition changed materially
Loading stability series	No reversion to pre-2020 level	Break is permanent within sample
Pre vs. post mean PC1 variance	15.9% → 19.8% (+3.9 pp)	Permanently elevated co-movement

6 Conclusion

6.1 Findings

1. *Low idiosyncratic explained variance. With cross-sectional standardisation, the leading PCs explain a modest fraction of variance (PC1: 4.7%, top-5: 16.6%). This confirms that removing the common market component leaves a rich, multi-factor idiosyncratic structure.*
2. *Style and sector factors. The four leading PCs correspond economically to:*
 - *a domestic cyclical vs. defensive growth axis,*
 - *a technology vs. commodities axis,*
 - *a green-energy vs. traditional finance axis, and*
 - *a PSU-industrial vs. consumer-finance axis.*
3. *Structural break at COVID-19. Rolling PCA, Chow test, and Procrustes alignment agree: the factor structure changed materially and permanently around March 2020. PC1 variance rose from a pre-COVID mean of 15.9% to 19.8% post-COVID, and the loading composition shifted by a Procrustes disparity of 0.782.*

6.2 Recommendations for Policymakers

- *Systemic risk monitoring. Rolling PC1 variance is a simple, model-free systemic risk indicator/a rising PC1 share signals increasing market-wide co-movement and reduced diversification benefits. SEBI or the Financial Stability Board could include such a measure in quarterly financial stability reports.*
- *Stress-testing correlation assumptions. Risk models calibrated on pre-2020*

correlation matrices will significantly underestimate tail dependence in a post-COVID regime. Regulators should require financial institutions to stress-test with post-2020 loading matrices.

- *Promoting sector diversification. The persistence of elevated market co-movement post-COVID argues for active policy support of sectors (domestic manufacturing, healthcare, digital infrastructure) that have low PC1 loadings and hence provide genuine portfolio diversification.*

6.3 Recommendations for Investors and Fund Managers

- *Dynamic factor exposure. Factor loadings are not stable; the Procrustes disparity of 0.782 confirms that a portfolio designed on pre-2020 factors provides very different exposures today. Risk models should be re-estimated at least annually.*
- *Use of rolling PC1 variance as a regime indicator. Elevated PC1 variance (above $\sim 20\%$) signals a high-correlation regime in which diversification is less effective; this is a natural trigger for reducing gross exposure or increasing hedging.*
- *Robust covariance estimation. Ledoit--Wolf shrinkage materially stabilises PCA-based factor estimates in rolling 24-month windows. Practitioners relying on raw sample covariance matrices are likely overfitting to noise in short windows.*

6.4 Limitations

The dataset is subject to survivorship bias (currently-listed stocks only). Rolling windows of 24 months cannot cleanly isolate the acute COVID period. Our economic interpretation of PCA loadings is based on stock names, not formal characteristics matching; corroboration via book-to-market and momentum data would strengthen these claims. Finally, the Chow test is conditional on March 2020 being specified a priori as the break date, rather than being

identified endogenously.

7 Data and Code Availability

Price data sourced from Yahoo Finance via yfinance 0.2.x (tickers: NSE listed stocks, suffix .NS). All analysis conducted in Python 3.11 with numpy 1.26.4, pandas 2.2.1, scikit-learn 1.4, statsmodels 0.14, and scipy 1.12. The full analysis notebook (stat4_pca_project_final.ipynb) is submitted alongside this report.

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